

# Derivatives of elementary functions



# Outline

Constant function and power functions

Derivatives of elementary trigonometric functions

Derivatives of the exponential and logarithmic functions with base  $e$

# Constant function and power functions

# Introduction: Constant function and power functions

- ▶ In what follows we derive the derivative functions of some of the main **elementary functions**.
- ▶ The tool is always the same: the **definition of derivative** as the limit of the difference quotient

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- ▶ We start from the constant function and the power functions  $x^n$  and  $x^\alpha$ : we will use the binomial theorem and the fundamental limits.

# Derivative of the constant function

## Observation 1

*Given the function  $f(x) = c$  with  $c \in \mathbb{R}$  constant, the derivative function is  $f'(x) = 0$ .*

# Proof

We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 0 = 0$$



# Derivative of the function $x^n$

## Observation 2

*Given the function  $f(x) = x^n$  with  $n \in \mathbb{N}, n \geq 1$ , the derivative function is  $f'(x) = n x^{n-1}$ .*

## Proof

From the binomial theorem (Newton's binomial formula) we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^n - x^n}{h} = \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\&= \frac{x^n + n x^{n-1} h + \sum_{k=2}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\&= n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0}\end{aligned}$$

hence

$$f'(x) = \lim_{h \rightarrow 0} n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0} = n x^{n-1}$$



## Proof: the case $n = 2$

With  $n = 2$ , we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$$



# Derivative of the function $x^\alpha$

## Observation 3

*Given the function  $f(x) = x^\alpha$  with  $\alpha \in \mathbb{R}$ , the derivative function is  $f'(x) = \alpha x^{\alpha-1}$  for  $x > 0$ .*

## Proof (1/2)

Let  $x > 0$ . We have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^\alpha - x^\alpha}{h} = \frac{\left(x \left(1 + \frac{h}{x}\right)\right)^\alpha - x^\alpha}{h} \\ &= x^\alpha \cdot \frac{\left(1 + \frac{h}{x}\right)^\alpha - 1}{h} \\ &\sim x^\alpha \cdot \frac{\alpha \frac{h}{x}}{h} = \alpha x^{\alpha-1} \quad \text{for } h \rightarrow 0\end{aligned}$$

where we used the fundamental limit

$$(1 + \varepsilon(h))^\alpha - 1 \sim \alpha \varepsilon(h) \quad \text{for } \varepsilon(h) \rightarrow 0$$

## Proof (2/2)

where

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \quad \text{for } h \rightarrow 0.$$

Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \alpha x^{\alpha-1} = \alpha x^{\alpha-1}$$



## Example: derivative function

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$$f(x) = x^{10} \qquad f'(x) = 10 x^9 \quad (\text{for } x \in \mathbb{R})$$

$$f(x) = \frac{1}{x} = x^{-1} \qquad f'(x) = -\frac{1}{x^2} \quad (\text{for } x > 0)$$

$$f(x) = \sqrt{x} = x^{\frac{1}{2}} \qquad f'(x) = \frac{1}{2\sqrt{x}} \quad (\text{for } x > 0)$$

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## Constant function and power functions:

- ▶  $f(x) = c \Rightarrow f'(x) = 0$
- ▶  $f(x) = x^n, n \in \mathbb{N}, n \geq 1 \Rightarrow f'(x) = n x^{n-1}$  (binomial theorem)
- ▶  $f(x) = x^\alpha, \alpha \in \mathbb{R} \Rightarrow f'(x) = \alpha x^{\alpha-1}$  for  $x > 0$  (fundamental limit of  $(1 + \varepsilon)^\alpha - 1$ )

# Derivatives of elementary trigonometric functions



# Introduction: Elementary trigonometric functions

- ▶ In this section we compute the derivatives of  $\sin x$  and  $\cos x$ .
- ▶ We will use the **addition formulas** and the fundamental limits

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2}, \quad \frac{\sin h}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0$$

- ▶ Trigonometric functions are the basis for describing periodic phenomena: knowing their derivatives is fundamental.

# Derivative of the sine function

## Observation 4

*Given the function  $f(x) = \sin x$ , the derivative function is  $f'(x) = \cos x$ .*

## Proof (1/3)

Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\sin(x+h) - \sin x}{h} \\&= \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\&= \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x\end{aligned}$$

Using the fundamental limit

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2} \quad \text{for } h \rightarrow 0$$

## Proof (2/3)

we have

$$\frac{\cos h - 1}{h} = h \cdot \left( \underbrace{-\frac{1 - \cos h}{h^2}}_{\rightarrow -\frac{1}{2} \text{ for } h \rightarrow 0} \right) \sim -\frac{1}{2} h \quad \text{for } h \rightarrow 0.$$

Using the fundamental limit

$$\frac{\sin h}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0$$

we have

$$\frac{\sin h}{h} \cos x \sim \cos x \quad \text{for } h \rightarrow 0.$$

## Proof (3/3)

Then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x \\ &= \lim_{h \rightarrow 0} \sin x \left( -\frac{1}{2} h \right) + \cos x = \cos x \end{aligned}$$



# Derivative of the cosine function

## Observation 5

*Given the function  $f(x) = \cos x$ , the derivative function is  $f'(x) = -\sin x$ .*

## Proof (1/2)

Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\cos(x+h) - \cos x}{h} \\ &= \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x\end{aligned}$$

## Proof (2/2)

Using the arguments of the previous proof, we have:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x \\ &= \lim_{h \rightarrow 0} \cos x \left( -\frac{1}{2} h \right) - \sin x = -\sin x \end{aligned}$$



# Summary

## Elementary trigonometric functions:

- ▶  $f(x) = \sin x \Rightarrow f'(x) = \cos x$
- ▶  $f(x) = \cos x \Rightarrow f'(x) = -\sin x$
- ▶ Tools: addition formulas and fundamental limits  $\frac{1-\cos h}{h^2} \rightarrow \frac{1}{2}$ ,  
 $\frac{\sin h}{h} \rightarrow 1$

# Derivatives of the exponential and logarithmic functions with base $e$



# Introduction: Exponential and logarithm with base $e$

- ▶ In this section we compute the derivatives of  $e^x$  and of  $\log x$ .
- ▶ We will use the fundamental limits

$$\frac{e^h - 1}{h} \rightarrow 1, \quad \log(1 + \varepsilon) \sim \varepsilon \quad \text{for } h, \varepsilon \rightarrow 0$$

- ▶ We will then apply the results to compute **tangent lines** to the graph of a function.

# Derivative of the exponential function

## Observation 6

*Given the function  $f(x) = e^x$ , the derivative function is  $f'(x) = e^x$ .*

# Proof

We have

$$\frac{f(x+h) - f(x)}{h} = \frac{e^{x+h} - e^x}{h} = e^x \cdot \frac{e^h - 1}{h} \sim e^x \quad \text{for } h \rightarrow 0$$

using the fundamental limit

$$\frac{e^h - 1}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0,$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} e^x = e^x$$



# Derivative of the logarithm function

## Observation 7

*Given the function  $f(x) = \log x$ , the derivative function is  $f'(x) = \frac{1}{x}$ .*

## Proof (1/2)

We have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\log(x+h) - \log x}{h} = \frac{\log\left(1 + \frac{h}{x}\right)}{h} \\ &\sim \frac{h}{x} \cdot \frac{1}{h} = \frac{1}{x} \quad \text{for } h \rightarrow 0\end{aligned}$$

## Proof (2/2)

We used the fundamental limit

$$\log(1 + \varepsilon(h)) \sim \varepsilon(h) \quad \text{for } \varepsilon(h) \rightarrow 0,$$

and

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \quad \text{for } h \rightarrow 0,$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{x} = \frac{1}{x}$$



## Example: tangent line (1/2)

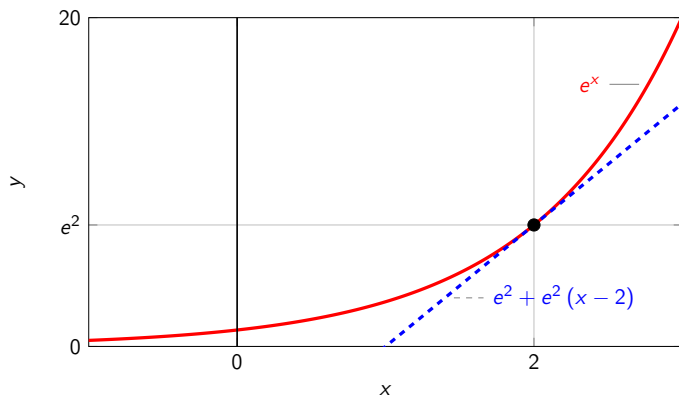
Let us compute the equation of the tangent line to the graph of the function

$$f(x) = e^x \text{ at the point with abscissa } x = 2$$

We have  $f(2) = e^2$ ,  $f'(x) = e^x$ ,  $f'(2) = e^2$ , hence the tangent line at the point  $(2, e^2)$  is:

$$y = f(2) + f'(2)(x - 2) = e^2 + e^2 (x - 2)$$

## Example: tangent line (2/2)



## Example: tangent line (1/2)

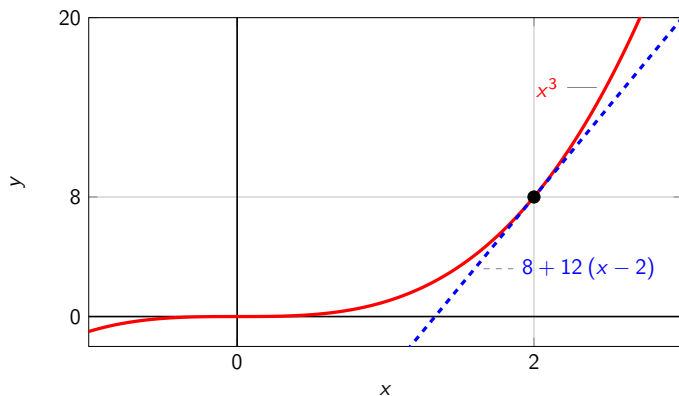
Let us compute the equation of the tangent line to the graph of the function

$$f(x) = x^3 \text{ at the point with abscissa } x = 2$$

We have  $f(2) = 8$ ,  $f'(x) = 3x^2$ ,  $f'(2) = 12$ , hence the tangent line at the point  $(2, 8)$  is:

$$y = f(2) + f'(2)(x - 2) = 8 + 12(x - 2)$$

## Example: tangent line (2/2)



# Summary

## Exponential and logarithm with base $e$ :

- ▶  $f(x) = e^x \Rightarrow f'(x) = e^x$
- ▶  $f(x) = \log x \Rightarrow f'(x) = \frac{1}{x}$
- ▶ Tangent line to the graph at the point with abscissa  $x_0$ :  
$$y = f(x_0) + f'(x_0)(x - x_0)$$

# Key messages

# Summary

## Derivatives of elementary functions:

| $f(x)$                               | $f'(x)$                                       |
|--------------------------------------|-----------------------------------------------|
| $c$                                  | $0$                                           |
| $x^n \ (n \in \mathbb{N}, n \geq 1)$ | $n x^{n-1}$                                   |
| $x^\alpha \ (\alpha \in \mathbb{R})$ | $\alpha x^{\alpha-1} \quad \text{for } x > 0$ |
| $\sin x$                             | $\cos x$                                      |
| $\cos x$                             | $-\sin x$                                     |
| $e^x$                                | $e^x$                                         |
| $\log x$                             | $\frac{1}{x}$                                 |

- All of them follow from the definition of derivative and from the fundamental limits.